Oscar Nolen
Charlotte, NC | 704-635-5160 | oscarnolen@gmail.com |
linkedin.com/in/oscar-nolen-846269153 | github.com/oscarno22

Experience

Software Engineer - Agentic AI, Ally Financial | Charlotte, NC | 01/2025–Present

Decedent Servicing Platform

- When customers pass away, Ally turns off their online services, applies business restrictions, and updates their profile. Lead engineer and primary production owner of the "Decedent" platform (FastAPI, Next.js) that streamlines this work for an operations team, projected to save ~$1M a year.
- Built daily claim ingestion (Salesforce) for the platform, plus a one-time 24-month backfill (~20K claims, ~1K new each month). The scheduler on FastAPI keeps every claim current with tiered refreshes based on claim status.
- The business wanted claims to load quickly, so the platform's dashboard employs indexing, server-side pagination, async batch processing on ECS, and Redis for caching and streaming updates to the UI in real time.
- Extended the platform with a LangGraph agent server (self-hosted RDS) that executes the servicing operations on each claim, using Textract for document analysis.

Agent platform

- Built the MCP Gateway for our business-facing agent platform: one central registry that resolves many MCP servers under a single host. Security is critical at a bank, so it supports dynamic Okta OAuth for seamless integration to Ally's authentication standard.
- The platform runs Conversational and Deep agents (LangGraph deepagents or AWS AgentCore), each in a fresh ECS task per session, with a skills registry and usage analytics.

Fraud recommendation engine (in development)

- Lead engineer for an event-driven fraud engine. It consumes fraud alerts from Kafka, orchestrates retrieval and caching of source data, computes feature flags over said data, triggers a Deep run on the agent platform to reason over those, then polls for the result and publishes a recommendation back to Kafka.

Across projects

- Own production releases end to end: Terraform, GitLab CI/CD, dependency and vulnerability fixes, load testing, APIs (Apigee and Kong), UAT, Dynatrace and CloudWatch monitoring, and the GenAI model approval process and documentation.
- Work with other teams and business lines to keep tabs on products and translate business needs into technical designs.

Software Dev Engineer Intern, Amazon Web Services | Seattle, WA | 05/2024–08/2024

- Contributed to AWS User Notifications (a notifications system) by designing an asynchronous central logging service that traces each notification through the data plane, so missing notifications and related issues can be identified.
- Built it with Lambda and DynamoDB, Kotlin APIs for retrieving notification traces, CDK infrastructure, CloudWatch monitoring, and a React frontend.

Software Dev Engineer Intern, Amazon Web Services | Seattle, WA | 05/2023–08/2023

- Contributed to Amazon Connect, an omnichannel cloud contact center, by building a reusable skeleton resource for replicating customer service workflows across accounts.
- Developed it in the existing control plane service with Java CRUD APIs and validation for its JSON representation, tested with JUnit and integration tests.

Amazon Technology Intern, Amazon Web Services | Matthews, NC | 05/2022–08/2022

- Worked on Babelfish for Aurora PostgreSQL, which lets Aurora understand commands written for Microsoft SQL Server. Identified catalog objects missing for Visual Studio, SSMS, and other SQL clients, and helped replace them through design reviews and code changes.

Projects

Market Data Aggregator | Rust, tokio, axum | github.com/oscarno22/market-aggregator

- An order book with one missed message is silently wrong, so this pipeline is built around data integrity. It ingests, normalizes, and fans out order books from three crypto exchanges, detecting gaps with sequence numbers and verifying book state with CRC32 checksums.
- Added sharding with a merging gateway, Prometheus metrics, a live SSE dashboard, and replay of recorded live sessions.

Tax Line AI | Python, AWS serverless | github.com/oscarno22/tax-line-ai

- Upload an invoice and an agent works out what each line item is and assigns its tax category. Built as an event-driven orchestration of Lambdas: API Gateway issues presigned S3 uploads, and each upload triggers a three-stage OpenAI Agents SDK agent (extract, classify, critic) backed by DynamoDB.
- Each stage fails gracefully with complete, partial, or failed states, status polling, and SNS email. Infra is in CloudFormation, deployed through GitHub Actions with keyless OIDC.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, Kotlin, C, C++
Backend and data: FastAPI, Kafka, Redis, PostgreSQL (Aurora RDS), DynamoDB, SQLite, REST APIs, event and batch processing

Cloud and DevOps: AWS (ECS, ECR, Lambda, S3, API Gateway, SNS, CloudWatch, Secrets Manager, Textract), Terraform, CloudFormation, CDK, Docker, GitLab CI/CD, GitHub Actions, Dynatrace
Frontend: React, Next.js, Node.js, Vite
Agentic AI: LangGraph, AWS AgentCore Harness, Model Context Protocol (MCP), OpenAI Agents SDK, Anthropic API, agentic coding (Claude Code), LLM Benchmarking
Machine learning: PyTorch, NumPy, Pandas, scikit-learn

Education

University of North Carolina at Charlotte | 08/2024–12/2025
Master of Science in Computer Science, concentration in AI, Robotics, Gaming

- Research: studied memorization in variational autoencoders. Wrote a custom VAE for retinal images from a diabetic retinopathy dataset and found that high memorization scores (above the 95th percentile) typically flag outliers from the rest of the dataset.

Duke University | 08/2020–05/2024
Bachelor of Science in Computer Science and Bachelor of Science in Mathematics